

HealthConnect Testing & Validation Report

Week 7 Technical Deliverable — Programmatic Dashboard Verification & Reassessment

DATASET Appointment Data	TOTAL RECORDS 5,000 Rows	ENVIRONMENT Python venv (Pandas)	STATUS PASSED (100%)
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1. Executive Summary & Objective

This validation report documents programmatic testing, data integrity audits, and segment-level consistency checks performed for the HealthConnect appointment analytics platform. Utilizing Python (pandas), raw records (HealthConnect_Appointment_Data.csv) were cross-verified against dashboard metrics.

Key Audit Finding

Programmatic calculations confirm 100% agreement with dashboard summary visual cards. Zero rounding discrepancies or logical drift were detected between the database and visual layers.

2. Data Quality & Structural Audit

Initial structural inspection flagged specific completeness gaps across secondary attributes. Pre-processing rules were verified to ensure missing attributes do not distort core KPI aggregations:

Attribute Name	Missing Count	Impact Level	Validation Action
reminder_channel	1,366	Moderate	Isolated nulls for channel effectiveness cross-tabs.
distance_to_clinic_km	90	Low	Excluded from distance mean calculations via dropna.
waiting_time_minutes	60	Low	Audited for wait-time outlier distribution.

3. Comprehensive Testing & Validation Record

Test ID	Validation Category	Programmatic Output (Python)	Dashboard Output	Status
VAL-01	Baseline KPI Volume	5,000 total records verified exact match	Total Appointment KPI card matches 5,000	PASS
VAL-02	Segmented Consistency	Cross-tabs confirm lead-time trends hold across subsets	Filtered slicer views maintain logical trends	PASS
VAL-03	Filter Responsiveness	appointment_type = General Consultation (n=3,255)	Slicer selection dynamically updates visuals	PASS
VAL-04	Anomaly Audit	Zero negative lead times; wait-time outliers flagged	Data range boundaries enforced correctly	PASS

4. Segmented Conclusion Testing

To ensure core insights are robust, conclusions were tested across operational segments. Segmented cross-tabulations confirm that the elevated no-show risk associated with extended booking lead times remains consistent across major appointment types and age brackets, proving findings are broadly generalizable.

5. Analytical Errors, Weaknesses & Limitations

Transparent evaluation of analytical constraints ensures high technical rigor:

- **Unimputed Missing Data Gaps:** Gaps in `reminder_channel` (1,366), `distance_to_clinic_km` (90), and `waiting_time_minutes` (60) were excluded list-wise rather than imputed, potentially introducing minor attrition bias.
- **Confounding Variables:** Lead-time correlations are treated as observational risk indicators rather than direct causal drivers, as unmeasured factors (e.g., patient health urgency) may influence behavior.
- **Sub-Segment Sample Variance:** Highly granular filter slices yield smaller sample sizes, which can cause percentage volatility in niche demographic brackets.

6. Reassessed Strategic Recommendations

Based on completed validation testing, segment checks, and analytical audits, the following evidence-based operational recommendations are established:

- **Lead-Time-Triggered Intervention Protocols:** Replace static reminder schedules with a tiered, dynamic engagement model. For appointments booked >14 days out, trigger automated check-ins at critical milestones (14 days, 7 days, and 48 hours prior) with interactive confirmation options.
- **Channel Effectiveness Optimization:** Standardize mandatory logging of communication channels during booking and pivot toward omni-channel automated workflows (SMS paired with WhatsApp or voice confirmation) tailored to high-risk service types.
- **Waiting Time & Distance Mitigation:** Establish real-time tracking for clinic wait times and integrate distance/travel factors into scheduling algorithms—offering telehealth alternatives for patients located further from the clinic (>10 km) to reduce transportation barriers.

7. Influence on the Wider HealthConnect Solution

The validated insights directly inform the architectural and workflow enhancements required for the broader HealthConnect platform:

- **Automated Risk-Based Scheduling:** Embed dynamic lead-time monitoring directly into the booking engine to auto-enroll long-lead appointments into high-touch reminder workflows.
- **Data Integrity & Mandatory Logging:** Enforce strict database validation rules for communication channels and outcome tracking to eliminate historical data completeness gaps.
- **Adaptive Patient Communications:** Transition from static clinic reminders to tailored, channel-optimized engagement based on service-level risk profiles and historical patient attendance.

8. Cross-Track Sharing (HC-POD Tracks)

Validated insights have been formally distributed to respective technical and operational HC-POD tracks:

- **Track 1 (Data Architecture & Engineering):** Enforce strict mandatory logging for `reminder_channel` and wait-time attributes during booking to resolve historical completeness gaps.
- **Track 2 (Predictive Modeling & Schedulers):** Integrate lead-time risk stratification and service-level weightings into the core scheduling optimization model.
- **Track 3 (Patient Engagement & Operations):** Deploy automated multi-touch reminder cadence (14-day, 7-day, 48-hour) and telehealth routing for high-distance cohorts.